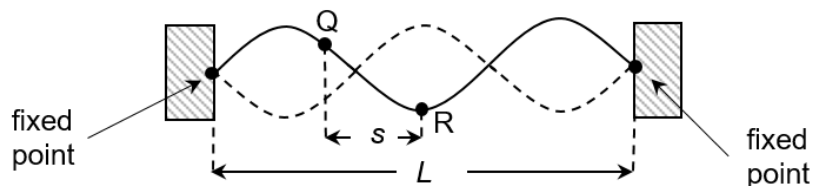


- 18** A stationary wave is formed on a stretched string of length L between two points as shown below.



Particles Q and R on the string are separated by distance s . The total energy of Q and of R are E_Q and E_R respectively.

Which of the following gives the correct phase difference between Q and R, and the relationship between E_Q and E_R ?

		phase difference / rad	total energy
	A	$\frac{2s}{L}\pi$	$E_Q = E_R$
	B	$\frac{2s}{L}\pi$	$E_Q < E_R$
	C	π	$E_Q = E_R$
	D	π	$E_Q < E_R$

Answer: D

Since Q and R are in adjacent segments of the same stationary wave, they are π radians apart.

As Q has a smaller amplitude compared to R, it has a smaller total energy.

19 Light of wavelength λ is incident normally on a diffraction grating of slit spacing five times the